

Factoring and chemistry on a simulated fault-tolerant quantum computer

Algorithms run end to end on a surface-code machine whose logical error rates were measured by circuit-level simulation

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[the explainer site](#) · github.com/Jaspersands/ft-algorithms ·

SUMMARY

Shor's algorithm (full modular exponentiation in Toffoli arithmetic, semiclassical approximate QFT) and iterative phase estimation of the H_2 and HeH^+ ground-state energies are run end to end on a simulated rotated-surface-code machine. Every logical error channel, idling, lattice-surgery merges and the surgery CNOT, was measured in 316 circuit-level experiments (0.82×10^9 shots) with stabilizer-qec under SD6 noise and extracted with baselines propagated through each operation; magic states come from published factory models. Whole surgery experiments predicted from their parts agree with circuit-level measurements to 0.93–1.26× (median 1.04×). At $p = 0.1\%$, factoring 15 reaches 90% of its noiseless peak probability at $d = 13$; phase estimation of H_2 needs 1.8×10^7 T gates per run. Extrapolated with the same error model, textbook-arithmetic RSA-2048 needs $d = 35$ and 32.5 M physical qubits for 1.96 years; Gidney's 2025 algorithm needs $d = 29$ under our error model where he assumed 25.

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1 Introduction

Resource estimates for fault-tolerant algorithms usually multiply a gate count by an assumed logical error rate. This work runs the algorithms instead. A logical-level simulator (ftsim, written for this project) executes the actual circuits, with every operation followed by its measured logical channel and every waiting patch accumulating measured idle error. The results are success probabilities as functions of code distance, physical error rate and magic-state source, with error budgets that say which operations cause the failures.

The approach rests on one fact and one approximation. The fact: under Pauli circuit noise, for a Clifford circuit decoded into a Pauli-frame correction, the logical action of one surface-code operation is exactly a Pauli channel, since every syndrome history ends in some logical Pauli. The approximation: consecutive operations' channels are composed as independent. Section 4.3 measures how good that is.

Contributions:

- A calibrated logical model of a rotated surface code under SD6 noise: idle (including logical Y), $Z \otimes Z$ and $X \otimes X$ merges and the surgery CNOT, from 316 experiments, with fits in d and p .
- A composition test: whole lattice-surgery experiments predicted from parts at the logical level and compared with circuit-level measurement.

- Shor’s algorithm with honest arithmetic (no use of the period anywhere in the circuit) run on that model for N up to 143, scored by peak probability because, for small N, uniformly random outcomes already “factor” with probability 1.00 (N = 15).
- Molecular phase estimation from first-principles Hamiltonians, to chemical accuracy.
- Extrapolation to RSA-2048 and FeMoco under the same error model, compared with published estimates.

2 Methods

2.1 Circuit-level calibration

All physical-level simulation uses stabilizer-qec 1.2 (the author’s engine, which matches Stim and PyMatching) with SD6 noise at $p \in \{0.1, 0.2, 0.3, 0.5\}\%$ and correlated matching. Experiments: memories of d, 2d, 3d and 4d rounds in both bases ($d = 3 \dots 11$), reference-qubit memories ($d \leq 7$), $Z \otimes Z$ and $X \otimes X$ merges of d rounds with d rounds apart before and after, the surgery CNOT (two merges and an ancilla patch, 4d rounds in all), and, for the composition test, three $Z \otimes Z$ measurements in a row and a three-patch merge. Each ran to 500 failures or a 300-second cap; every record keeps its seed.

Channels through fidelities. A distribution over flip patterns has Pauli fidelities $f(s) = \sum_x P(x)(-1)^{s \cdot x}$, which multiply under composition. The per-round idle channel is the slope of $\log f$ over the memory lengths (preparation and readout cancel). An operation’s own channel is the measured one divided by its baseline, where the baseline (preparation plus d rounds before, d rounds plus readout after, each the square root of a 2d-round memory’s fidelity) is first pushed through the ideal operation: a CNOT copies an X error on the control onto the target and a Z on the target onto the control, and an error before a merge changes its outcome. Dividing naively gives negative probabilities for “X on the control only”. Uncertainties are bootstrap (300 resamples).

Logical Y. X- and Z-basis memories see XuY and ZuY . A reference-qubit experiment gives the full channel: the patch is projected noiselessly (every Z check by a noiseless Pauli-product measurement, which round 1 is compared with), entangled with a perfect qubit by a noiseless $Z_R Z_L$ measurement, run, and ended by a perfect round and noiseless $X_R X_L$ and $Z_R Z_L$ measurements. Without the initial projection an X error before the first Z-check readout flips $Z_R Z_L$ unseen.

Fits. Each component $\epsilon(d, p)$ is fitted per p as $\log \epsilon = a + b(d + 1)/2$ by weighted least squares over points with at least ten events, and globally as $A(p/p*)^{((d+1)/2)}$. Values beyond the largest measured distance are extrapolations and are marked as such in every figure.

2.2 The machine

operation	rounds	logical channel
X, Y, Z	0	none (Pauli frame)
H	0	none: transversal, orientation tracked in the fast block †
S, S†	d	$Z \otimes Z$ merge channel (
T, T†	d	merge channel + Z w.p. ϵ_T † + wrong outcome → Z w.p. $\frac{1}{2}$ (twirled S)
CX, CZ	2d	measured surgery-CNOT channel, idling included
CCX, CCZ	2d	3 merges + Z-type w.p. ϵ_{CCZ} † + wrong outcomes → twirled CZ; d rounds idle
M, R	1	one round of idling
idle	per round	measured (p_X, p_Y, p_Z)
feed-forward	10	decoder reaction time before a TABLE †

† marks cited inputs. Magic states: injection ($\epsilon_T = p$, conservative against Li 2015’s $\approx 0.4p$), 15-to-1 distillation (Litinski 2019, Table 1: 4.5×10^{-8} per T and 5.2×10^{-11} per CCZ at $p = 0.1\%$) and cultivation (Gidney, Shutty and Jones 2024: 2×10^{-9} at $p = 0.1\%$, 4×10^{-11} at 0.05%; $|CCZ\rangle$ by 8T-to-CCZ, $28 \epsilon_T^2$). Physical qubits: Litinski’s fast block, $2n + \lceil \sqrt{8n} \rceil + 1$ tiles of $2(d + 1)^2$ qubits, plus factories sized by their published qubit-rounds per state. One round is 1 μ s.

The noise compiler gives every qubit a clock. An operation starts when its qubits are free (a TABLE also waits for its records plus the reaction time); each waiting qubit receives the composed idle channel for its gap; the operation’s channel follows it with a class label for attribution. REPEAT blocks are barriers, scheduled once; TABLE cases are padded to the longest, so the schedule does not depend on outcomes. Channel composition is written with log1p/expm1: per-round rates reach 10^{-20} at large d and a naive $1 - (1 - 2q)^n$ cancels to zero.

2.3 The simulator

ftsim (Rust) runs programs in a Stim-like format: Clifford + T + Toffoli gates, Pauli channels on one to three qubits, measurements, REPEAT, and TABLE (classical feed-forward choosing a block by earlier records). Backends: a sparse state (hash map of basis states; runs of permutation and phase gates fused per basis state), which handles 36-qubit Shor circuits in milliseconds per shot, and a dense vector. Each shot's RNG derives from (seed, shot), so results do not depend on threads.

Stratified sampling. With K the number of static faults, $P(\text{success}) = \sum_k P(K = k) \cdot S_k$. $P(K = k)$ is computed exactly (a truncated generating polynomial over the site tree, REPEAT by exponentiation), and S_k is estimated from shots with exactly k faults, drawn by sampling k sites with weights $q/(1 - q)$ and rejecting duplicates, which is exactly the conditional law. Strata beyond the tail cutoff carry their mass as an interval. The one-fault stratum, with faults logged, gives each operation class's harm.

2.4 Algorithms

Shor. Cuccaro ripple-carry adders; modular addition of a classical constant with a sign flag (Vedral–Barenco–Ekert, Beauregard); controlled modular multiplication with uncomputation by the inverse and a controlled swap; $3n + 6$ qubits and $2n(2n(10n + 12) + n)$ Toffolis. One control qubit is recycled through $2n$ rounds of a semiclassical approximate QFT whose correction rotations, chosen by TABLE from the previous $\lceil \log_2 2n \rceil + 2$ records, are synthesized by gridsynth to $\epsilon = 10^{-3}/2n$ and verified as matrices. The base is $a = 2$ for every N . Every multiplier is the generic circuit, even when its constant is 1.

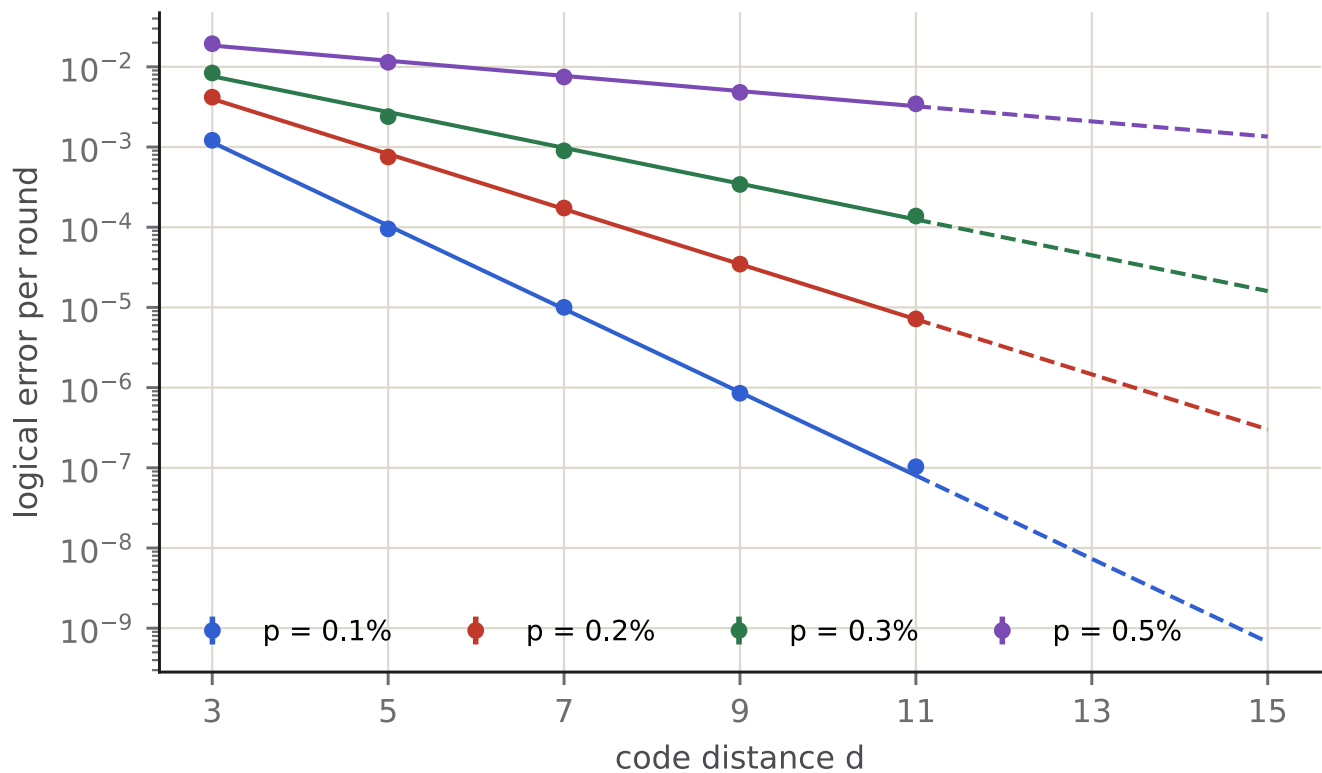
Chemistry. STO-3G integrals of s Gaussians in closed form (Boys F_0), restricted Hartree–Fock with DIIS, Jordan–Wigner with interleaved spins, and Z_2 tapering of both spin parities (Bravyi et al.). Iterative phase estimation reads 10 bits of $U = \exp(-i(H - E_{\text{HF}})\tau)$ with $\tau = 2\pi$, approximated by four fourth-order Suzuki steps; controlled Pauli rotations use only uncontrolled synthesized R_z 's, so their global phases stay global.

3 Results

3.1 The calibrated machine

316 experiments, 0.82×10^9 shots and 459,686 logical failures. Suppression factors Λ (per two units of distance):

p	idle (XuY)	idle (ZuY)	CNOT X on target	CNOT Z on control	Z $\otimes$ Z patch Z	Z $\otimes$ Z wrong outcome
0.1%	10.93	10.06	7.91	7.10	7.12	12.30
0.2%	4.86	4.72	3.42	3.43	3.17	6.33
0.3%	2.80	2.85	2.01	2.01	1.98	2.97
0.5%	1.55	1.62	1.13	1.11	1.11	1.58

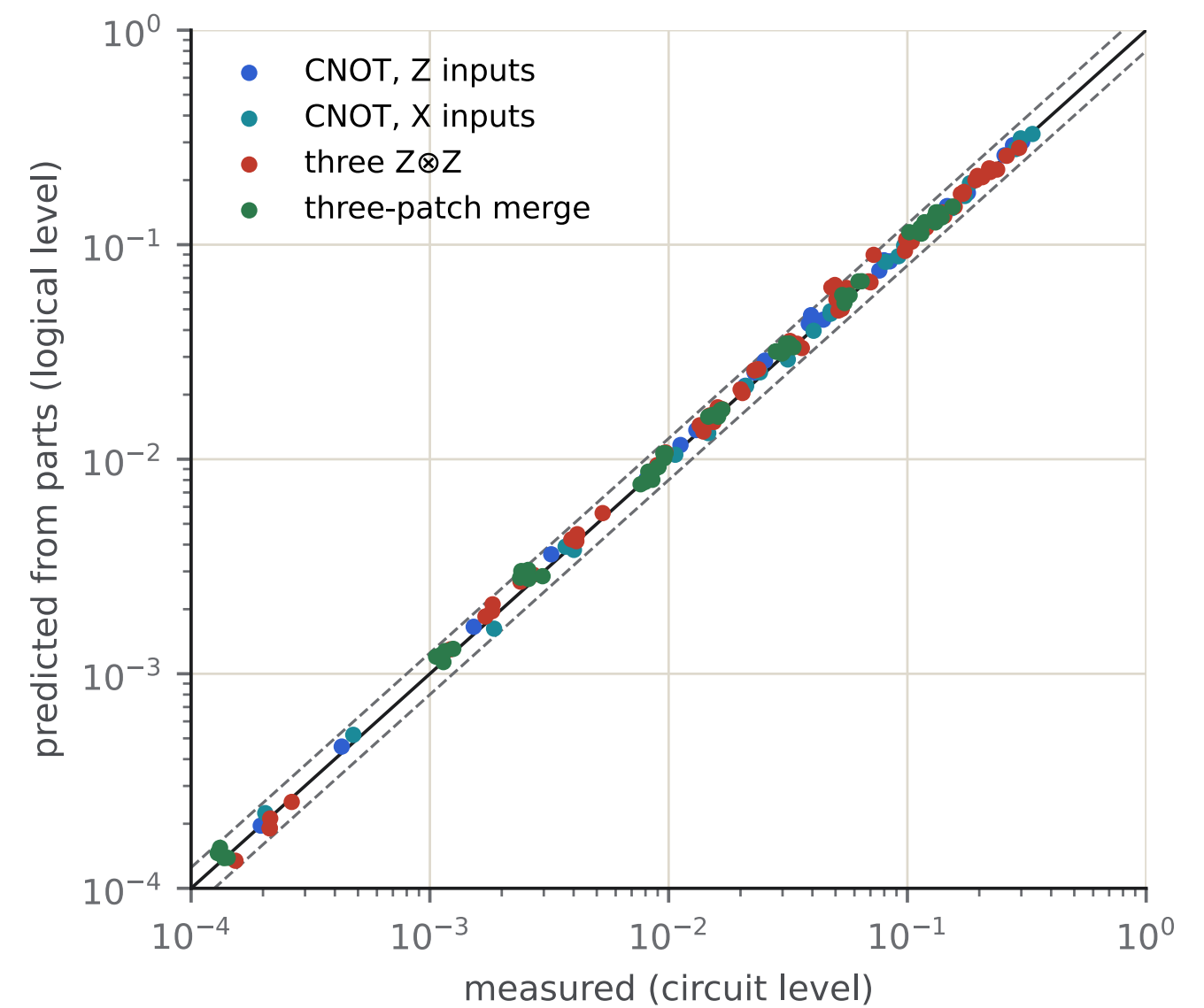


Per-round logical error of an idle patch (XuY). Points measured (1σ), lines per- p fits, dashed beyond the largest measured distance.

At $p = 0.1\%$ an idle patch errs 8.8×10^{-7} per round at $d = 9$, and the fit gives 4.3×10^{-15} at $d = 25$. Logical Y is 4% of the XuY rate at $d = 3$, 43 $\times$ what independent X and Z would give: Y errors are included as measured.

Cross-check. 36 experiments rerun with Stim and PyMatching (correlated) agree with stabilizer-qec within statistics (largest $|z| = 2.2$).

3.2 Composition



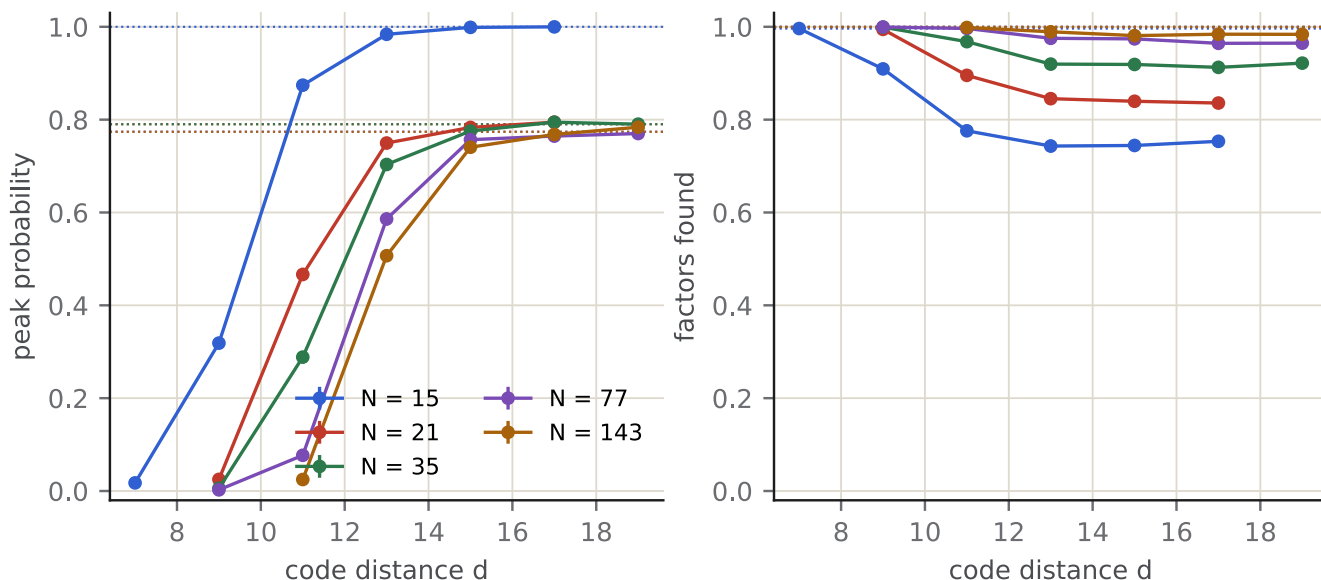
Observable failure rates of whole surgery experiments: predicted from calibrated parts by ftsim vs measured at circuit level; dashed $\pm 25\%$.

Across 48 experiment-noise-distance combinations the predicted failure rate is 0.93–1.26× the measured one, median 1.04×. The independent-composition model is accurate to about 10% and leans pessimistic.

3.3 Shor’s algorithm

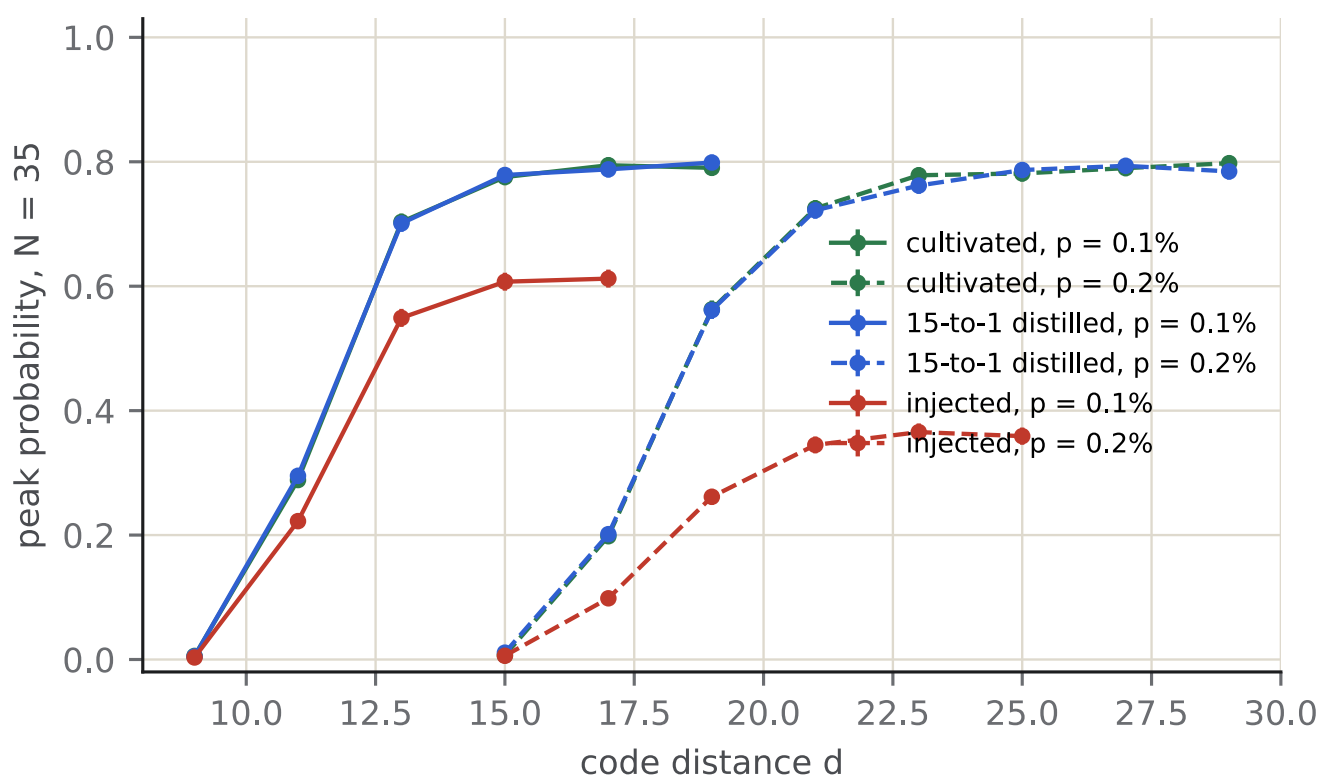
At $p = 0.1\%$ with cultivated magic states:

N	n	logical qubits	Toffolis	T	noiseless peak	random peak	random factors	d for 90%	physical qubits	run time
15	4	18	3,360	223	1.000	0.0156	0.996	13	55.4 k	225 ms
21	5	21	6,250	317	0.790	0.0059	0.999	13	58.1 k	418 ms
35	6	24	10,440	425	0.790	0.0029	1.000	15	63.4 k	808 ms
77	7	27	16,170	519	0.774	0.0018	1.000	15	67 k	1.25 s
143	8	30	23,680	635	0.774	0.0009	1.000	15	70.8 k	1.81 s



Left: peak probability vs d (cultivated states, $p = 0.1\%$); dotted: noiseless. Right: the same runs scored by factors found; dotted: random-outcome baseline.

Factoring success is a poor score for small N. Classical post-processing that tries the convergents' small multiples finds the factors from most outcomes, and the noisiest runs look best. The peak probability falls to its random baseline ($r/2^m$) instead.



N = 35: peak probability vs d for each magic-state source; solid $p = 0.1\%$, dashed $p = 0.2\%$.

magic states	p	ϵ_T †	ϵ_{CCZ} †	best peak	d for 90%
cultivated	0.1%	2.0×10^{-9}	1.1×10^{-16}	0.794 (d = 17)	15
cultivated	0.2%	10.0×10^{-8}	2.8×10^{-13}	0.798 (d = 29)	21
15-to-1 distilled	0.1%	4.5×10^{-8}	5.2×10^{-11}	0.799 (d = 19)	15
15-to-1 distilled	0.2%	3.6×10^{-7}	3.3×10^{-9}	0.793 (d = 27)	21
injected	0.1%	1.0×10^{-3}	2.8×10^{-5}	0.612 (d = 17)	never
injected	0.2%	2.0×10^{-3}	1.1×10^{-4}	0.366 (d = 23)	never

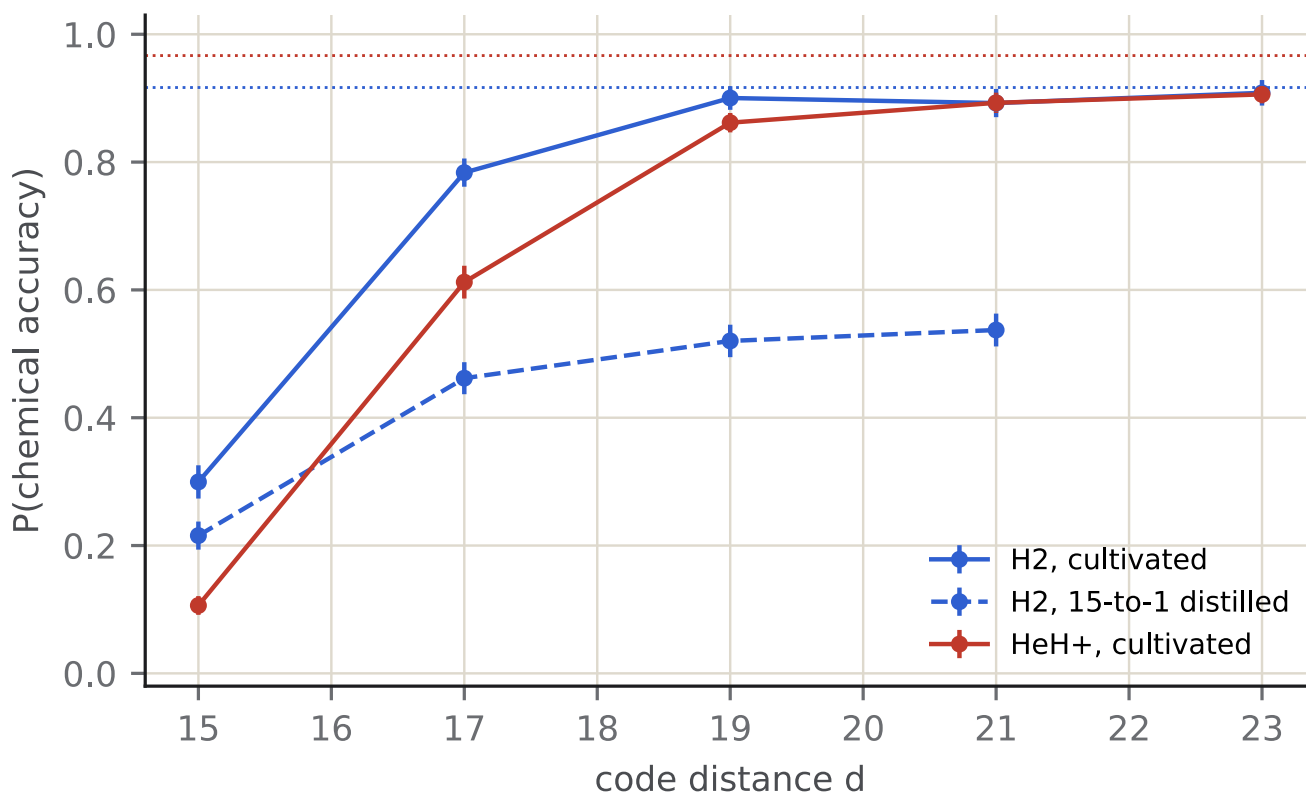
Error budget (N = 35, d = 15, cultivated, p = 0.1%):

class	expected faults	share	score of one-fault runs
idle	0.027	80.3%	0.394
CNOT	5.3×10^{-3}	15.9%	0.564
CCZ	1.2×10^{-3}	3.7%	0.667
T	1.7×10^{-5}	0.0%	–
S	1.0×10^{-5}	0.0%	–
measure	3.7×10^{-8}	0.0%	–

Idling accounts for 80% of the expected faults. The textbook circuit runs one Toffoli at a time (the schedule averages 5.13·d rounds per Toffoli with 3.2 patches busy) while every other patch waits.

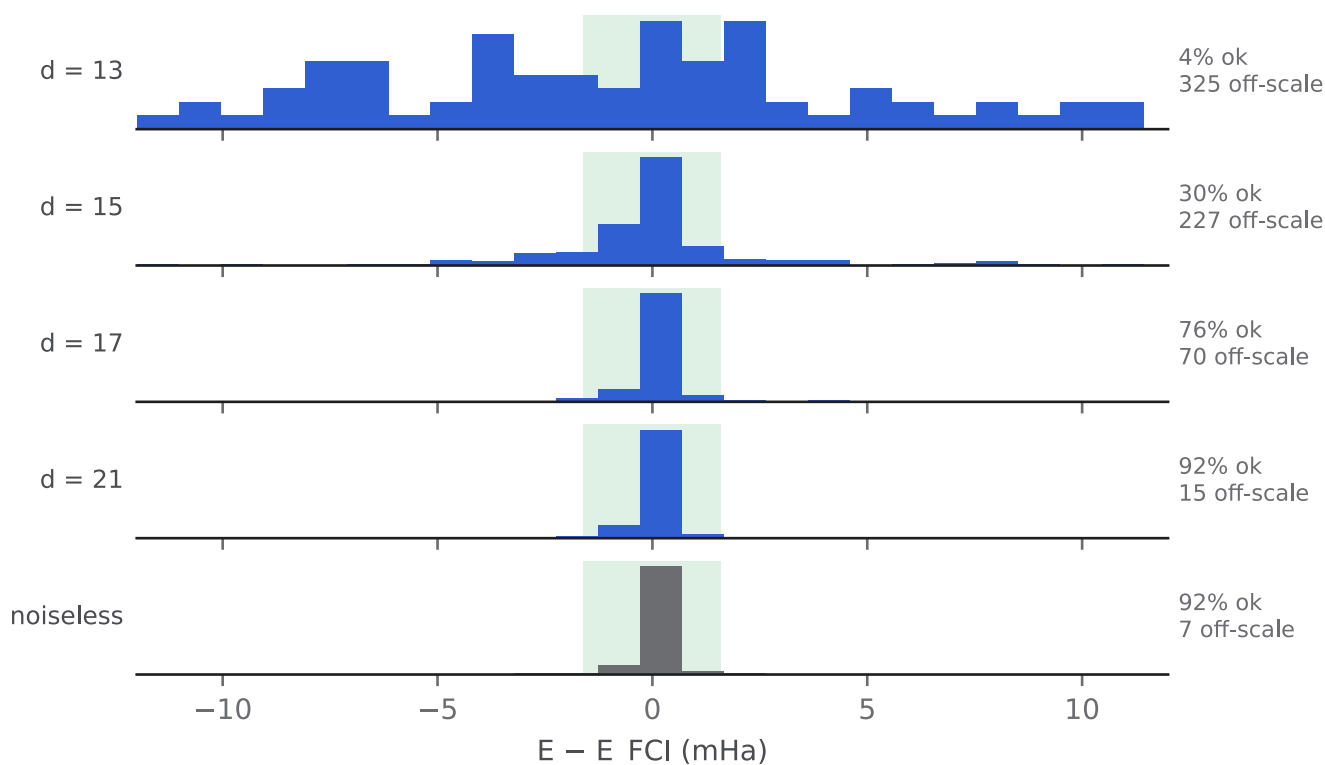
3.4 Phase estimation

molecule	logical qubits	terms	T gates	E_HF (Ha)	E_FCI (Ha)	noiseless	best	d for 90%	run time	physical qubits
H2	3	5	1.8×10^7	-1.116714	-1.137276	0.92	0.91 (d = 23)	19	7.2 min	22.1 k
HeH+	3	9	4.4×10^7	-2.841836	-2.851466	0.97	0.91 (d = 23)	21	20.3 min	22.4 k



Probability of chemical accuracy (1.6 mHa from FCI) vs d at $p = 0.1\%$; dotted: noiseless.

Each run uses $\approx 1.8 \times 10^7$ T gates for H_2 , so the magic-state error sets a floor that no distance removes. With cultivated states (2×10^{-9}) that floor is a few percent; with 15-to-1 distilled states (4.5×10^{-8}) most runs fail.

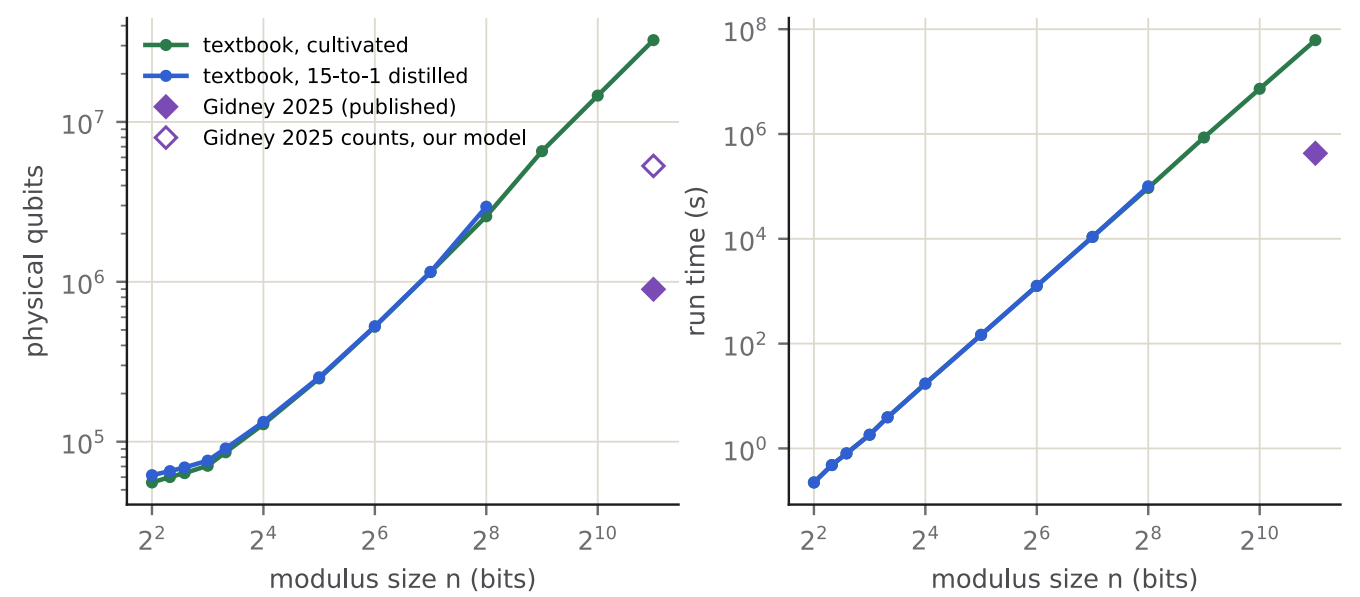


The answer itself: energies returned by 400 individual H_2 runs at several distances (cultivated states, $p = 0.1\%$), as the error from the exact energy; shaded: chemical accuracy; grey: the same circuit without noise.

3.5 To scale

Operation counts of the textbook circuit are exact; the schedule's shape (rounds per Toffoli in units of d , 5.13; CNOTs per Toffoli, 2.28; patches busy, 3.18) is fitted on compiled instances and is stable across them. The smallest odd d with at most 0.1 expected faults:

workload	logical qubits	Toffolis	published d	published qubits	our d	our qubits	run time
RSA-2048, textbook arithmetic	6,150	3.4×10^{11}	—	—	35	32.5 M	1.96 years
RSA-2048, Gidney 2025	1,409	6.5×10^9	25	898 k	29	5.31 M	4.63 days
FeMoco (THC), Lee et al. 2021	2,142	5.3×10^9	31	≈ 4 M	29	8.01 M	2.45 days



Textbook Shor from the simulated sizes to RSA-2048 ($p = 0.1\%$).

Under our measured error model, Gidney's 2025 RSA-2048 algorithm needs $d = 29$ where he assumed 25: our idle rate at $d = 25$ is 4.3×10^{-15} per round against his 10^{-15} . Our physical-qubit totals use plain patches throughout and so bound his yoked-storage design from above. The textbook arithmetic's 3.4×10^{11} Toffolis exceed what 15-to-1 distillation's 5.2×10^{11} can support beyond $n = 256$.

4 Validation

- `ftsim`: dense = sparse on random circuits; 100 random circuits equal Qiskit's statevector to 10^{-10} ; noisy distributions equal Aer's exact density matrices (χ^2 , plain and stratified); WebAssembly = native.
- Arithmetic verified on every input by an independent bit-level simulator; Shor's noiseless outcomes match the exact distribution; synthesized rotations verified as matrices; QPE matches its exact Trotterized distribution.
- Chemistry: integrals, HF and FCI equal PySCF to 10^{-8} Ha; Szabo and Ostlund's H_2 reproduced; tapering preserves the spectrum.
- Calibration: Stim + PyMatching cross-check; reference-qubit experiment against plain memories; composition test.

5 Limitations

- Composition is accurate to about 10% (Section 4.2), measured only for $d \leq 7$.
- Most algorithm results lie beyond the directly measured distances ($d \leq 9$ for the CNOT at $p = 0.1\%$): extrapolated along measured exponential fits, marked in the figures.
- Magic-state error rates and footprints, the H and S operation model and the timing constants are cited inputs, not measured here.
- The arithmetic is textbook and sequential. Modern constructions (windowed arithmetic, measurement-based uncomputation, parallel Toffolis) would cut the cost by orders of magnitude; the scaling section separates the error model from the algorithm by also

evaluating Gidney's 2025 counts.

6 Reproduction

```
tools/build.sh                # the engine (Rust → Python module)
python tools/calibrate.py      # circuit-level experiments (resumable)
python tools/fit_model.py      # data/calibration/model.json
python tools/compose_check.py; python tools/xcheck.py
python tools/run_shor.py; python tools/run_qpe.py; python tools/scale.py
python tools/site_data.py; python tools/report.py
```